

$$\operatorname{Re}(\langle u, v \rangle) = \frac{\|u+v\|^2 - \|u-v\|^2}{4}$$

$$\|u+v\|^2 = \langle u+v, u+v \rangle \stackrel{\text{SL2}}{=} \langle u+v, u \rangle + \langle u+v, v \rangle \stackrel{(3)}{=} \langle u, u \rangle + \langle v, u \rangle + \langle u, v \rangle + \langle v, v \rangle$$

$$\begin{aligned} &\stackrel{\text{SS}}{=} \langle u, u \rangle + \langle v, v \rangle \\ &\quad + \overline{\langle u, v \rangle} + \langle u, v \rangle \\ &= \langle u, u \rangle + \langle v, v \rangle + 2\operatorname{Re}(\langle u, v \rangle) \\ &= \|u\|^2 + \|v\|^2 + 2\operatorname{Re}(\langle u, v \rangle) \end{aligned}$$